Heart Failure and Arrhythmias

NURA 340 Pathophysiology | Lecture Study Guide

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Lecture roadmap: Heart failure (systolic vs. diastolic, compensation, left vs. right) → Cardiomyopathies (dilated, hypertrophic, restrictive) → Dysrhythmias (sinus bradycardia, sinus tachycardia, atrial fibrillation, atrial flutter)

This lecture brings together the structural, ischemic, and inflammatory concepts covered earlier and asks what happens when the heart can no longer sustain adequate cardiac output.

$$CO = SV \times HR$$

Heart failure and cardiomyopathy impair stroke volume through lost contractility, stiffened ventricles, or remodeling. Dysrhythmias impair heart rate or eliminate the coordination of contraction needed to fill the ventricles or eject blood. As you study each condition, watch how fluid retention, electrical shifts, and acid-base disturbances interact with these mechanical and electrical failures.

Exam note: EKG strips are shown to help you visualize rhythms, but they will not appear on the test.

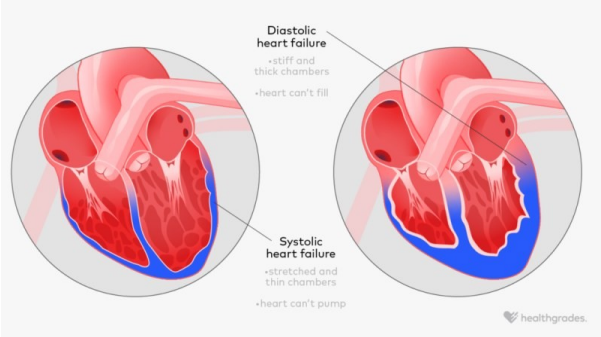
Heart Failure

Heart failure is a complex clinical syndrome from any structural or functional impairment of ventricular filling or ejection of blood. What matters most is getting blood out to the body: if the ventricles cannot eject, or are not filling enough, cardiac output suffers.

Functional Classification: Systolic vs. Diastolic

This classification asks: is the problem filling, or getting the blood out?

	Systolic (HFrEF)	Diastolic (HFpEF)
Ejection fraction	Reduced: ≤ 40%	Preserved: ≥ 50%
Deficit	Pumping / squeezing	Filling / relaxation
Walls	Thin, stretched, dilated	Stiff, thick, hypertrophied (or fibrotic)
What happens	Inotropic impairment: the myocardium loses contractility. Ventricles dilate, wall tension rises, and the chambers cannot squeeze blood out effectively during systole	The squeeze stays normal, but the ventricles cannot relax during diastole (when most filling occurs). Pressure rises, filling is harder, end-diastolic volume falls, and stroke volume drops
Most common cause	Ischemic heart disease (prior MI) is the most common cause of heart failure overall	Long-standing uncontrolled systemic hypertension



Diastolic vs. systolic heart failure (lecture slide)

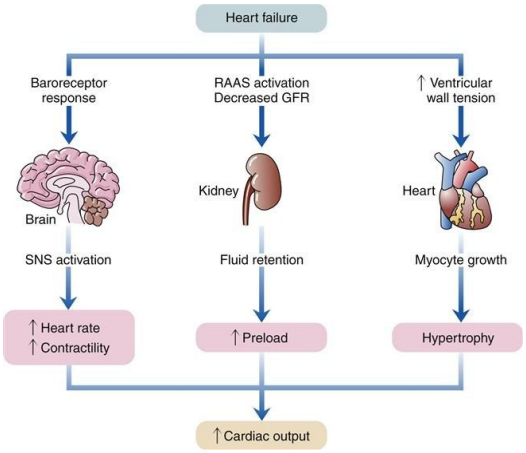
Neurohormonal Compensatory Mechanisms

When cardiac output falls, the body mistakes poor pumping for systemic hypovolemia and triggers three compensatory cascades, the same ones seen in hypovolemia. They help in the short term, but in heart failure there is no true volume deficit, so over time they hurt.

System	What it does	Effect on the heart
Sympathetic nervous system	Baroreceptors sense falling arterial pressure; epinephrine and norepinephrine are released	↑ heart rate and contractility (↑ myocardial oxygen demand) and peripheral vasoconstriction (↑ afterload)
RAAS	Decreased renal perfusion stimulates renin, leading to angiotensin I then angiotensin II	Angiotensin II: potent vasoconstriction (↑ afterload) and fibroblast proliferation (remodeling). Aldosterone: sodium and water reabsorption (↑ preload) while wasting potassium and magnesium
ADH (vasopressin)	Posterior pituitary responds to arterial underfilling	Free water retention in the renal collecting ducts

The result: increased afterload (the heart must push harder against resistance) and increased preload (it must handle more volume), when it is already struggling. Eventually it can no longer compensate.

Dilutional hyponatremia: Chronic RAAS and ADH activation can expand total body water beyond sodium retention. This usually signals a long-standing process: the patient is chronic, not well controlled, and not doing well with maintenance.



Neurohormonal compensation in heart failure (lecture slide)

Left-Sided vs. Right-Sided

Left vs. right tells you where the primary failure is. Systolic vs. diastolic tells you how the muscle is failing. Etiologies and risk factors are the same for both sides.

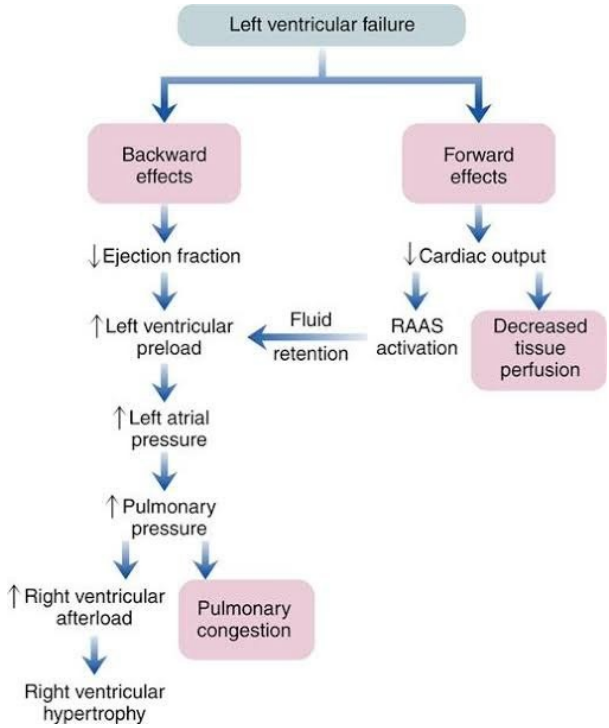
- **Causes:** ischemic heart disease and prior MI (most common overall); long-standing uncontrolled hypertension (most common cause of diastolic HF); valvular heart disease; dilated or hypertrophic cardiomyopathy; metabolic disorders (diabetes, thyroid disease, kidney failure).
- **Risk factors:** advanced age, hypertension, CAD, myocardial injury, diabetes, metabolic syndrome, high sodium intake, tobacco, alcohol abuse, cardiotoxic medications such as chemotherapy.

Many of these are linked in both directions (did heart failure cause the kidney disease, or the reverse?). Heart failure at baseline already shows these signs; fluid retention makes them more prominent. People live with HF for a long time if they follow the plan of care, adhere to medications, watch lifestyle factors, and watch for worsening signs and changes in fluid status. Signs that worsen steadily near the end of life are managed for comfort, not to treat the HF itself.

Left-Sided Heart Failure

The big one: it directly affects respiratory status and systemic perfusion, and the left is the larger ventricle. Blood backs up from the left ventricle into the left atrium, then the pulmonary veins and capillaries. Elevated pulmonary capillary hydrostatic pressure forces fluid into the alveoli. It can continue backward into the right side, which is how left-sided failure often causes right-sided failure. Even a small amount of fluid overload can hurt these patients.

Backward effects: lungs, then right ventricle and atrium. **Forward effects:** decreased cardiac output, so decreased tissue perfusion.



Backward and forward effects of left ventricular failure (lecture slide)

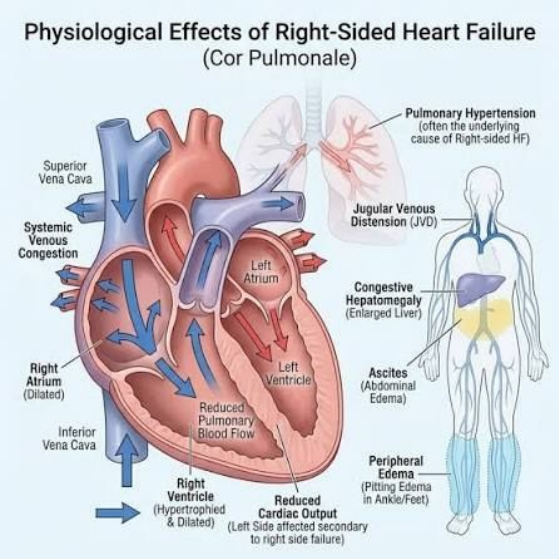
Signs and symptoms

- **Dyspnea on exertion**
- **Orthopnea:** shortness of breath lying down.
- **Paroxysmal nocturnal dyspnea:** sudden nighttime awakening, gasping for air.
- **Pulmonary crackles; pink, frothy sputum** (pulmonary edema).
- **S3 gallop:** heard in early diastole during rapid passive filling of noncompliant, fluid-overloaded ventricles.
- **Systemic:** fatigue and weakness, sweating, nausea, chest pain, arrhythmias, cold clammy skin, loss of appetite (the body is not getting the blood and nutrients it needs). If the right side fails, right-sided signs appear too.

Right-Sided Heart Failure

The right ventricle (systolic or diastolic failure) cannot move blood to the pulmonary circuit, so blood backs up into the right atrium and the systemic venous circulation. Most often it is caused by left-sided failure ("hand in hand"). It can also be caused by chronic lung disease: the resulting pulmonary hypertension makes the right ventricle work harder against high pressure to push blood into the pulmonary artery.

Thicker is not stronger: Over time the walls thicken, but unlike gym muscle, a larger cardiac muscle is stiffer, fills with less volume, and has a harder time contracting.



Effects of right-sided heart failure (lecture slide)

Signs and symptoms

- **Jugular vein distension (JVD)**
- **Hepatomegaly, splenomegaly, ascites:** fluid accumulates behind the liver, spleen, and abdomen from increased venous pressure.
- **Dependent peripheral edema**
- **Weight gain:** rapid gain signals fluid overload the heart cannot keep up with.
- **Other:** bloating, loss of appetite, fatigue and weakness, chest pain, palpitations.

Diagnostics

Test	Purpose / finding
Echocardiogram (gold standard)	Differentiates systolic from diastolic HF; measures chamber dimensions, stroke volume, and ejection fraction
BNP	Released by cardiac muscle in response to wall stretch and volume overload. < 100 helps rule out heart failure as the cause of acute dyspnea; > 400 confirms cardiac volume overload (the heart cannot keep up with its volume)
Electrolytes and renal function	The body diverts blood to vital organs and the kidney is not one of them, so check for kidney damage. Potassium and sodium imbalances can also impair the heart muscle
Chest X-ray	Cardiomegaly, pulmonary vascular congestion, pleural effusions

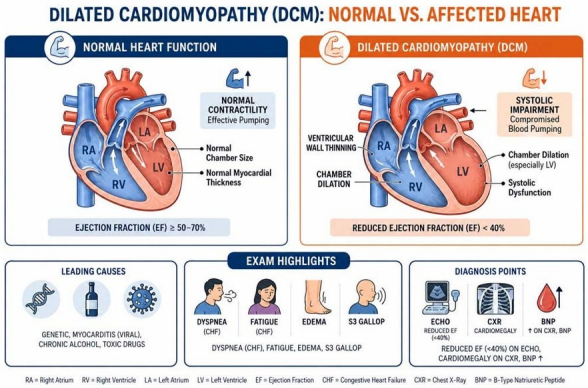
Complications and impact: high rates of illness and mortality, left atrial dysfunction, AFib, pulmonary hypertension, progression from left- to right-sided failure, exercise intolerance, and fatigue. Heart failure greatly impairs daily activities and causes emotional strain.

Cardiomyopathies

A heterogeneous group of myocardial diseases with mechanical or electrical dysfunction leading to inappropriate ventricular hypertrophy or dilation.

	Dilated	Hypertrophic	Restrictive
Structure	Enlarged chambers, thin weak walls	Thickened septum and walls	Rigid, noncompliant walls (normal thickness)
Result	Weak squeeze (systolic)	Outflow obstruction; impaired filling (diastolic)	Impaired passive filling (diastolic)
Frequency			Least common

Dilated Cardiomyopathy (DCM)

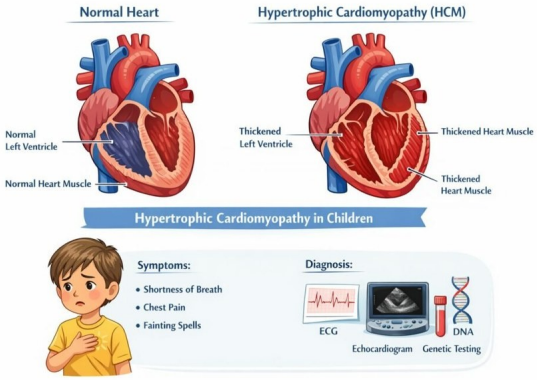


Normal heart vs. dilated cardiomyopathy (lecture slide)

Usually biventricular: both ventricles dilate, and the walls become thinned, stretched, and attenuated. Severe systolic dysfunction (reduced ejection fraction), sometimes with diastolic dysfunction, valve leakage, and wall stress. Compensatory mechanisms worsen remodeling and myocardial demand, so it progresses to heart failure.

- **Etiology:** genetic mutations; viral myocarditis; chronic alcoholism; MI (damaged tissue, less contractility); high blood pressure (increased afterload dilates the ventricle over time); or unknown.
- **Clinical:** often asymptomatic early, then shortness of breath, cough, chest pain, palpitations, dizziness, sudden weight gain; S3 gallop and systolic HF symptoms as it progresses.
- **Diagnostics:** echo (reduced EF, dilated ventricles); chest X-ray (enlarged heart); BNP (increased stretch); EKG (may show low voltage or conduction blocks, since the stretched, thin walls house the conduction system).
- **Complications:** heart failure, arrhythmias, stroke, blood clots, kidney failure, sudden cardiac death.

Hypertrophic Cardiomyopathy (HCM)

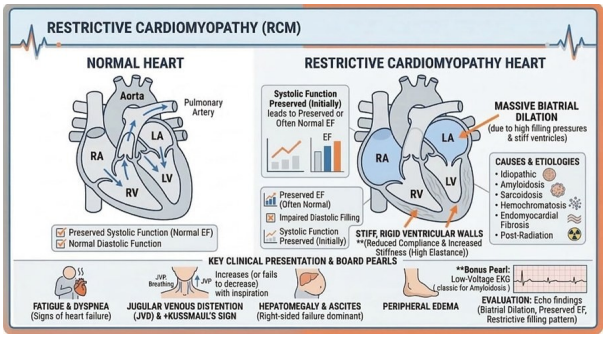


Normal heart vs. hypertrophic cardiomyopathy (lecture slide)

Asymmetric septal hypertrophy with disarray of myocardial fibers. The thickened septum between the ventricles blocks blood leaving through the aortic valve (dynamic left ventricular outflow tract obstruction), and the thick, non-relaxing walls cause severe diastolic dysfunction with impaired passive filling. Bigger muscle does not mean a bigger squeeze.

- **Etiology:** autosomal dominant genetic mutation; can start at any age but is most often diagnosed in adults in their 30s or 40s.
- **Sudden cardiac death:** the leading identifiable cause of sudden cardiac death in young people. A young athlete with no cardiac history collapses in cardiac arrest and is diagnosed with HCM at the hospital. A bystander providing basic life support until an advanced team arrives is critical.
- **Clinical:** often asymptomatic; may have exertional dyspnea, fatigue, angina, syncope during exertion, or sudden cardiac death.
- **Diagnostics:** echo (gold standard) shows asymmetric septal thickness. EKG may show left ventricular hypertrophy, but reading a 12-lead takes specialized skill beyond most bedside nurses.
- **Complications:** atrial fibrillation to sudden cardiac arrest.

Restrictive Cardiomyopathy (RCM)



Normal heart vs. restrictive cardiomyopathy (lecture slide)

Normal wall thickness but rigid, noncompliant ventricular walls from fibrotic or infiltrative deposits, so the ventricles cannot relax and fill. Blood backs up into the atria, lungs, and systemic veins. Systolic function is normal early on; this is truly a diastolic problem.

What to know: You do NOT need to memorize amyloidosis, sarcoidosis, or hemochromatosis. Understand that inflammatory or infiltrative deposits make the walls rigid and noncompliant.

- **Etiology (rare):** amyloidosis (abnormal protein fibers build up in organs; may cause carpal tunnel syndrome), sarcoidosis (granulomas; lung changes on imaging), hemochromatosis (too much iron stored; bronze skin, liver damage, joint pain, diabetes), radiation fibrosis.
- **Clinical:** right-sided HF dominant: JVD, S4 gallop; extra symptoms depend on the cause.
- **Diagnostics:** echo shows normal ejection fraction with biatrial enlargement (large atria) and normal ventricle sizes; biopsy confirms infiltration.
- **Prognosis:** poorest of all cardiomyopathies; average survival is only 2–5 years after diagnosis. The least common, but the highest mortality.

Cardiac Dysrhythmias

Cardiac rhythm depends on automaticity, excitability, and conduction velocity. When structural stretch, ischemia, or autonomic surges disrupt the electrical pathways, dysrhythmias develop. This lecture covers four of the most common and generally non-lethal ones (the lethal rhythms requiring CPR come later in the program): sinus bradycardia, sinus tachycardia, atrial fibrillation, and atrial flutter.

Sinus Node Dysrhythmias

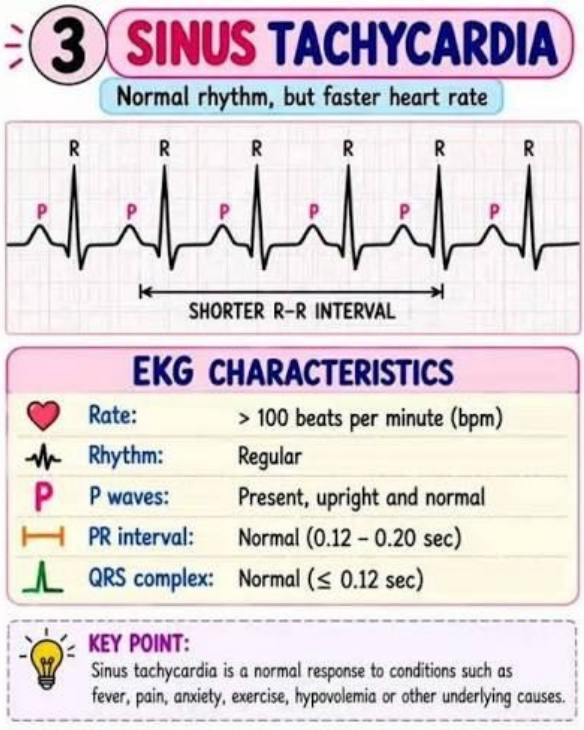
In sinus rhythms the electrical pathway is fully intact: SA node → AV node → bundle of His → Purkinje fibers. The rate is just slow, fast, or normal. There is a P wave before every QRS with normal intervals. Whether a sinus rate is a concern depends on cardiac output, the patient's signs and symptoms, and what the patient's normal is.

	Sinus bradycardia	Sinus tachycardia
Rate (adult)	< 60 bpm	> 100 bpm (slides: > 101)
Origin / consequence	SA node; low rate drops cardiac output	SA node; shortened diastolic filling and ischemia risk
Risk factors / causes	Athletes (efficient, conditioned hearts); sick sinus syndrome; inferior wall MI (ischemia to the SA node artery); hypothyroidism; hyperkalemia; medications	Fever; dehydration / hypovolemia (compensatory); hypoxia; pain; anxiety; pulmonary embolism; hyperthyroidism; anemia; sympathetic agents (caffeine, cocaine, medications)
Signs and symptoms	Asymptomatic (fine if baseline, e.g., athletes); lightheadedness, dizziness, syncope, fatigue, hypotension, altered mental status (decreased perfusion)	Palpitations, chest tightness, shortness of breath, lightheadedness, plus signs of the underlying cause



Sinus bradycardia (lecture slide)

- **Bradycardia and output:** if heart rate is low but stroke volume is adequate, the patient tolerates it. The concern is when stroke volume is inadequate or the rate is so low (e.g., 30 bpm) that cardiac output per minute cannot perfuse the body. Few patients tolerate that, and only briefly.
- **Tachycardia and output:** a typical adult upper limit is about 160–180 bpm. As the rate climbs beyond roughly 120–140, diastolic filling time shortens, stroke volume falls, and cardiac output drops, while myocardial oxygen demand rises: a loop. Sinus tachycardia is not dangerous in itself; find and treat the cause. If the patient is hemodynamically stable (good BP, mental status, perfusion), concern is lower. When those are affected, intervene quickly.



Sinus tachycardia (lecture slide)

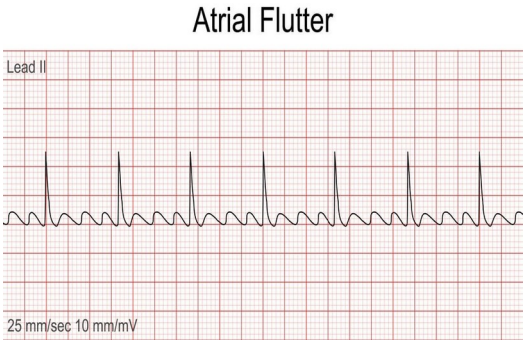
Atrial Dysrhythmias

Atrial fibrillation and atrial flutter are among the most common dysrhythmias that do not follow the normal conduction pathway. Many people live with them as their normal rhythm outside the hospital. A chaotic or rapid reentry circuit takes over from normal SA node firing. The risk factors, concerns, and often treatment are the same, but they are different rhythms.

	Atrial fibrillation (AFib)	Atrial flutter
Mechanism	Multiple chaotic reentry microcircuits firing continuously, about 300–600/min, from all over the atria (uncoordinated macro/micro ectopic foci). The atria quiver instead of contracting together	A single macro-reentrant circuit looping continuously, typically around the tricuspid valve area of the right atrium, about 250–350/min
EKG	Fibrillatory waves; absence of P waves; irregularly irregular; QRS tall and narrow	"Sawtooth" flutter waves (not true P waves); regular or predictably irregular ventricular rate
Analogy	A silent dance party: everyone has headphones on, dancing to a different beat	A city of nightclubs: each club's dancers move together, but the clubs are not coordinated
Risk factors	Hypertension, CAD, heart failure, mitral valve disease, hyperthyroidism, advanced age, obesity, obstructive sleep apnea, heavy acute alcohol intake	Similar to AFib; especially common with COPD, pulmonary embolism, and structural heart disease
Hemodynamic concern	Loss of atrial kick; thromboembolism and stroke risk	Partial loss of atrial kick; thromboembolism and stroke risk



Atrial fibrillation (lecture slide)



Atrial flutter (lecture slide)

Why Afib and Flutter Are Dangerous

- **The AV node gatekeeper:** not all 250–600 atrial impulses reach the ventricles. The AV node's prolonged refractory period (driven by parasympathetic tone) filters them, so the ventricular rate

can be normal or fast. Tall, narrow QRS complexes mean the ventricles are squeezing well, which matters most for cardiac output.

- **Loss of atrial kick:** the active atrial contraction in late diastole that pushes the last 20–30% of ventricular filling is lost. That blood pools in the atria, creating a high risk of clots.
- **Thromboembolism and stroke:** a clot can move to the ventricle and be pushed out to the body, causing ischemic injury. Watch for signs of stroke.
- **AFib with rapid ventricular response (RVR):** if the AV node conducts faster because of high sympathetic tone, the ventricular rate rises: increased heart rate, decreased stroke volume, decreased cardiac output. This is more dangerous.

Diagnostics for Dysrhythmias

- **12-lead EKG (first line):** identifies the rhythm, its origin, wave morphology, conduction intervals, and ischemic changes.
- **Ambulatory (Holter) monitoring:** worn from 24 hours to several weeks at home to capture intermittent dysrhythmias.
- **Serum electrolytes:** potassium, calcium, and magnesium can enhance automaticity (ectopic atrial beats, ventricular dysrhythmias) or slow conduction.
- **Thyroid function tests:** when no other cause is found for a sinus bradycardia or tachycardia.

Quick Review

Condition	Key points
Systolic HF	EF ≤ 40%; pumping deficit; dilated thin walls; most commonly from ischemic heart disease
Diastolic HF	EF ≥ 50%; filling deficit; stiff thick walls; most commonly from long-standing hypertension
Left-sided HF	Backs up into the lungs: DOE, orthopnea, PND, crackles, pink frothy sputum, S3
Right-sided HF	Backs up into the body: JVD, hepatosplenomegaly, ascites, dependent edema, weight gain; most often caused by left-sided HF
BNP	< 100 rules out HF as cause of acute dyspnea; > 400 confirms cardiac volume overload
DCM / HCM / RCM	Dilated: weak squeeze, low EF. Hypertrophic: thick septum, outflow obstruction, young-athlete sudden death. Restrictive: rigid walls, biatrial enlargement, poorest prognosis
Sinus brady / tachy	Rate < 60 / > 100 with normal conduction; worry only if perfusion or cardiac output suffers
AFib / flutter	Loss of atrial kick, clots, stroke; AFib: irregularly irregular, no P waves; flutter: sawtooth

Notes on Transcript Cleanup

Corrected mis-transcriptions: "R**S" is RAAS; "converting angiotensin to angiotensin I" is angiotensinogen to angiotensin I; "proximal nocturnal dyspnea" is paroxysmal; "jugular vein ascension" is distension; "eccentric electrical gatekeeper" is read as "electrical gatekeeper"; "human cancer unit" is read as a later hematology/oncology unit. The lecture says sinus tachycardia is above 100, while the slides say > 101; I listed > 100 with the slide value noted. The remark that diastolic heart failure patients have complications was read as applying to heart failure generally. The line about the SA node "failing to fire just one impulse" was garbled, so I described it as a chaotic reentry circuit taking over.